

BioGen NL Agent – Phase 1 and Phase 2 Report

Data Parsing, Indexing, Search and Factually Grounded RAG

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1 Introduction

In high-stakes settings such as biomedical question answering, factual grounding is essential. This report covers two phases of the BioGen NL Agent: Phase 1 builds and evaluates the retrieval layer over a PubMed corpus, and Phase 2 extends it into a factually grounded RAG pipeline with LLM-as-a-judge evaluation.

2 BioGen

2.1 Data Parsing, Indexing and Search

2.1.1 Corpus and Topic Data

Each PubMed record was parsed into three fields: `pmid`, `title`, and `abstract`. Each topic specifies an information need via `topic`, `question`, and `narrative`, which were concatenated into a single query string for retrieval.

2.1.2 Relevance Judgments

Relevance judgments were derived from the provided submission annotations. For each question, cited PMIDs in accurate system answers were inspected. Citations labeled as *supporting* were assigned relevance score 2, citations labeled as *neutral* were assigned relevance score 1, and citations labeled as *contradicting* or *not relevant* assigned relevance score 0. This produced graded relevance judgments suitable for NDCG evaluation, as well as binary relevance sets for precision, recall, MAP, and MRR.

2.1.3 Indexing in OpenSearch

The retrieval system was implemented in OpenSearch. Each document was indexed with identity fields (`pmid`, `title`, `abstract`), three separate lexical content fields each using a different simi-

larity function (BM25, JM, Dirichlet), and a dense **embedding** field for KNN search. Separating the lexical fields enables a fair comparison over the same document text.

2.1.4 Retrieval Models

We implemented four retrieval strategies: **BM25**, **Language Model with Jelinek-Mercer smoothing (JM)**, **Language Model with Dirichlet smoothing (LMD)**, and **KNN** dense retrieval. For KNN, documents and queries were encoded into dense vectors using a sentence embedding model and retrieved by nearest-neighbor search in OpenSearch.

2.2 LLM Augmented Generation Phase 2

2.2.1 Overview

In Phase 2, the retrieval component developed in Phase 1 was extended into a factually grounded retrieval-augmented generation pipeline. The objective of this phase is to generate a single biomedical answer for each topic using only evidence extracted from the retrieved PubMed documents.

Since KNN was the strongest retrieval method in Phase 1, the Phase 2 system uses the KNN test run as input. For each query, the top 3 retrieved documents are selected, and up to 3 reference sentences are extracted per document.

2.2.2 Reference Sentence Selection

Each abstract was split into sentences; very short fragments were discarded. Each sentence was scored by pairing it with the query using `ncbi/MedCPT-Cross-Encoder`, and the top 3 sentences per article were kept as reference evidence. Results were saved to `phase2_reference_sentences.json`.

2.2.3 Answer Generation

The selected reference sentences were passed to an instruction-tuned biomedical decoder model, `google/medgemma-4b-it`. The prompt instructed the model to answer only using the provided reference sentences and to avoid adding external biomedical knowledge.

The prompt constrained the model to at most 250 words, required every sentence to carry at least one citation in `[PMID:id]` format (max three per sentence, only from the provided reference set), and instructed the model to acknowledge insufficient evidence rather than hallucinate.

The generated answers were saved to `phase2_generated_answers.json`.

2.2.4 LLM-as-a-Judge Evaluation

The final stage used GPT-4o through the IAedu API as an LLM-as-a-judge. Two types of judgments were collected.

Each reference sentence was labeled for alignment (**Required**, **Borderline**, **Unnecessary**, **Inappropriate**), and each generated answer was labeled for entailment (**Supported**, **Partially Supported**, **Unsupported**).

The judge results were saved to `phase2_judge_results.json`.

Prompt calibration. The specification prompts were adjusted before running the full evaluation: the judge’s role was changed from “expert annotator” to “expert biomedical annotator”, and explicit input labels (**Question:**, **Complete answer:**, **Reference sentences:**) were added to prevent the model from confusing which section it was evaluating. A small-sample test confirmed that labels matched expectations before the full run.

2.2.5 BERT Embeddings and Self-Attention Exercise

In addition to the RAG pipeline, we implemented the required BERT visualization exercise using `bert-base-uncased`. Three analyses were performed.

Positional embeddings. To isolate positional information, we built a 200-token sequence by repeating the word “medicine” and projected the positional embeddings to 2D with PCA. The PCA plot traces a smooth spiral (Figure 1, left), showing that nearby positions cluster together and distance grows gradually along the curve. To confirm this, we also computed the full 200×200 pairwise distance matrix (right). The dark diagonal shows that neighboring positions are similar; the bright yellow corners show that positions far apart in the sequence are maximally different. The smooth color gradient between them confirms there are no sudden jumps — distance grows steadily with sequence separation.

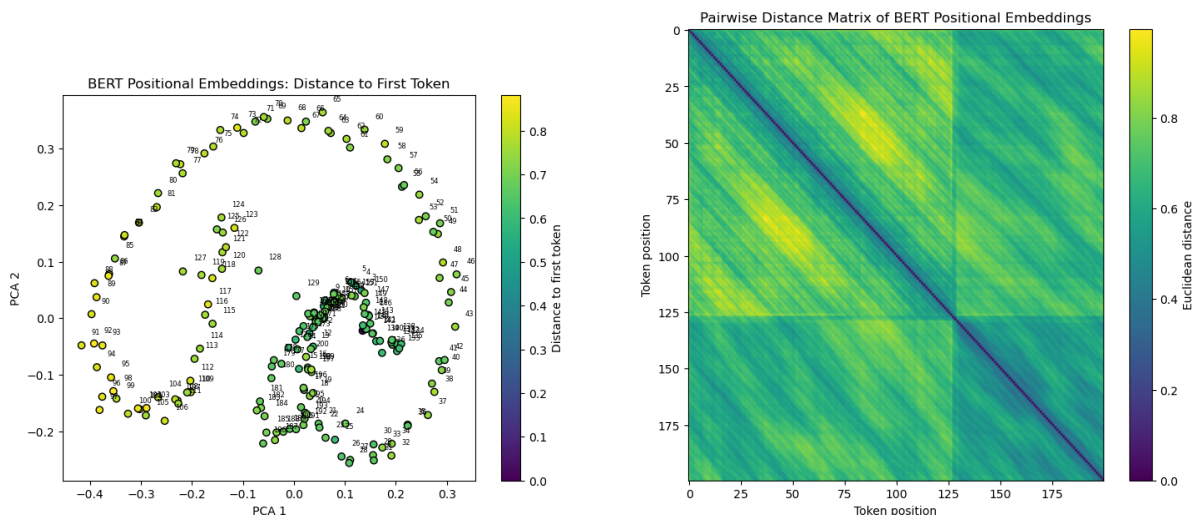


Figure 1: Left: PCA of BERT positional embeddings colored by distance to position 1. Right: pairwise distance matrix. Both confirm continuous, ordered position encoding.

Contextual embeddings. We fed BERT a query-passage pair: “How can sleep apnea be improved?” paired with “Sleep apnea can be improved by weight loss, avoiding alcohol, using

positional therapy, and applying continuous positive airway pressure.” The two sentences are joined by a [SEP] token, giving a 37-token input. We extracted the token representations at each of the 12 transformer layers and projected them with PCA.

Figure 2 shows each layer side by side. At layer 0 tokens are tightly packed, since the model has not yet processed any context. By layers 9–11 the spread is roughly ten times larger, and tokens that appear in both sentences (such as *sleep* and *apnea*) have diverged into distinct positions — shaped by their different surrounding context in the query versus the passage. This is how the cross-encoder builds a representation that reflects meaning, not just word identity.

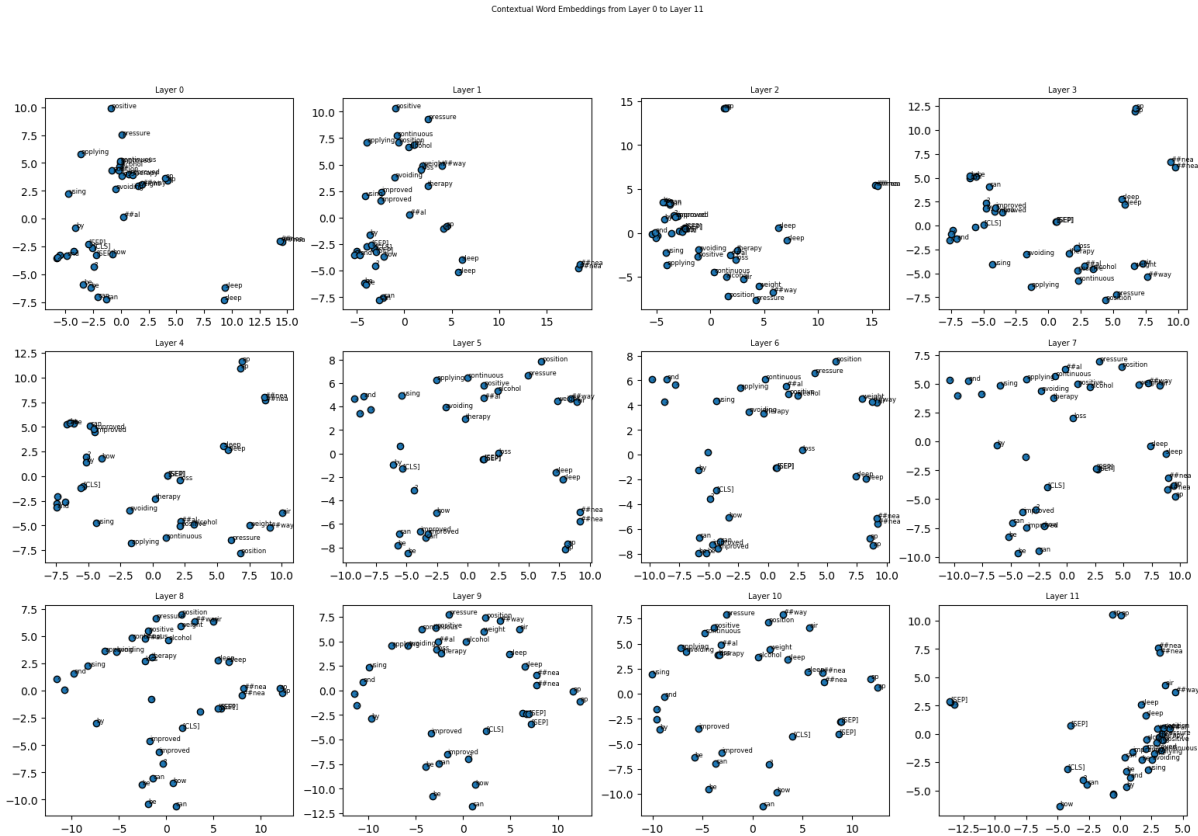


Figure 2: PCA projections of BERT token embeddings at each of the 12 transformer layers

Self-attention maps. Finally, we visualized the self-attention patterns of all 12 heads in layer 1. Each cell (i, j) shows how much word i attends to word j . Figure 3 shows all 12 heads at once.

The heads specialize in distinct roles. Heads 1 and 4 show a diagonal — each word attends only to itself, preserving token identity in early layers before deeper context builds up. Heads 5 and 9 route every word to [CLS], which the cross-encoder uses as the final relevance score. Head 10 is the most relevant for retrieval: query words attend directly to passage words, comparing question to evidence in a single forward pass — something a bi-encoder cannot do. Head 11 links the same word across sentences, connecting *sleep* in the query to *sleep* in the passage. Together with the positional and contextual analyses, these patterns show how BERT builds from ordered token positions, through layer-by-layer context, to cross-sentence relevance judgments.

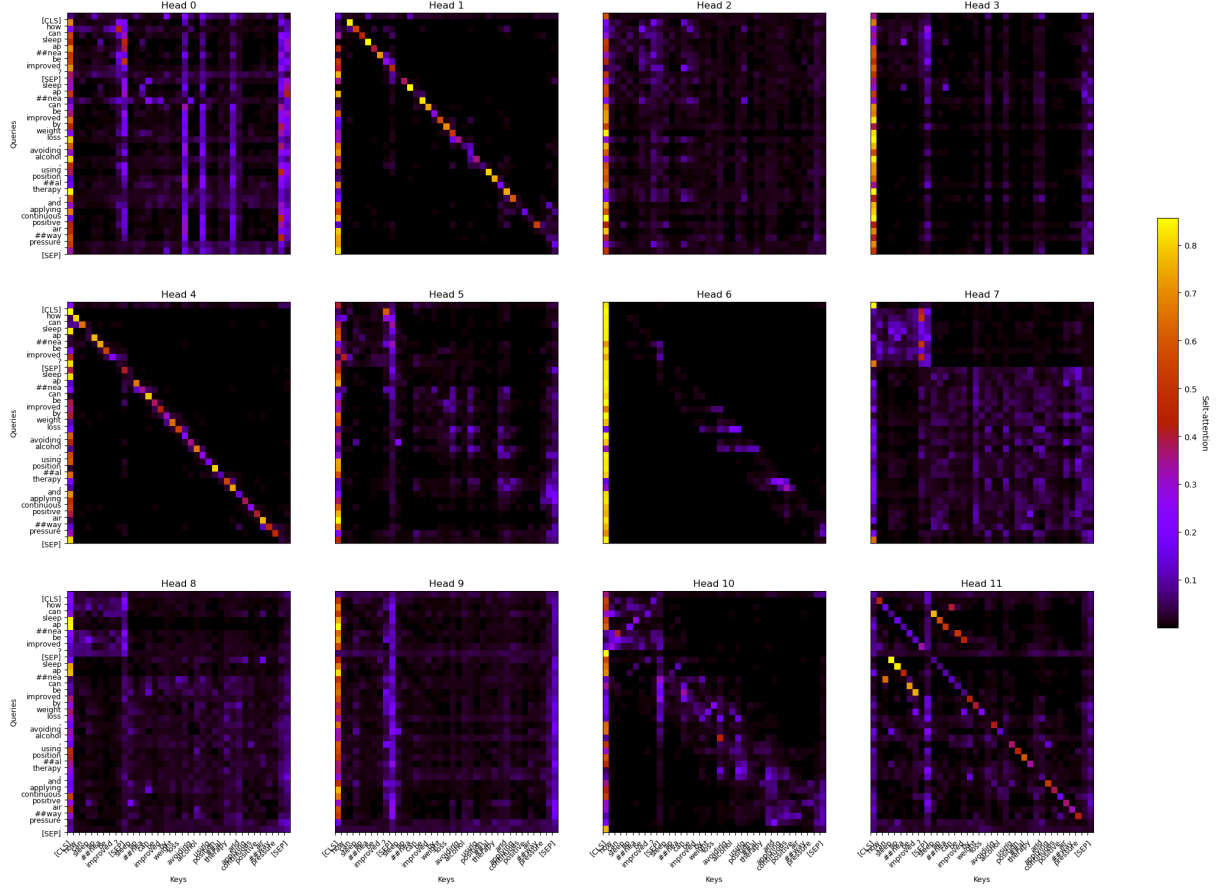


Figure 3: Self-attention matrices for all 12 heads in layer 1

3 Evaluation

3.1 Experimental Setup

The experiments use a PubMed corpus of 4,194 documents and 33 test queries with graded relevance judgments. Models were assessed with P@10, R@100, NDCG@10, NDCG@100, MAP, and MRR; precision-recall curves were also generated. All reported results are on the test split.

3.2 Results and Discussion

3.2.1 Main Quantitative Results

Table 1 presents the main retrieval results on the test set.

Table 1: Phase 1 test set retrieval results.

Method	P@10	R@100	NDCG@10	NDCG@100	MAP
BM25	0.6667	0.8435	0.6417	0.7652	0.5979
JM	0.6758	0.8080	0.6503	0.7404	0.5693
LMD	0.6636	0.8104	0.6315	0.7374	0.5659
KNN	0.7485	0.8946	0.7207	0.8247	0.6719

3.2.2 Discussion

KNN achieved the best results across all metrics, capturing semantic similarity even when lexical overlap is limited. Among lexical methods, BM25 was the strongest baseline with competitive NDCG and the highest MRR. JM and LMD were weaker under default settings but improved after parameter tuning (Section 3.3). KNN is therefore the best candidate for Phase 2.

3.2.3 Precision-Recall Analysis

Figure 4 includes the precision-recall curves of the four retrieval methods.

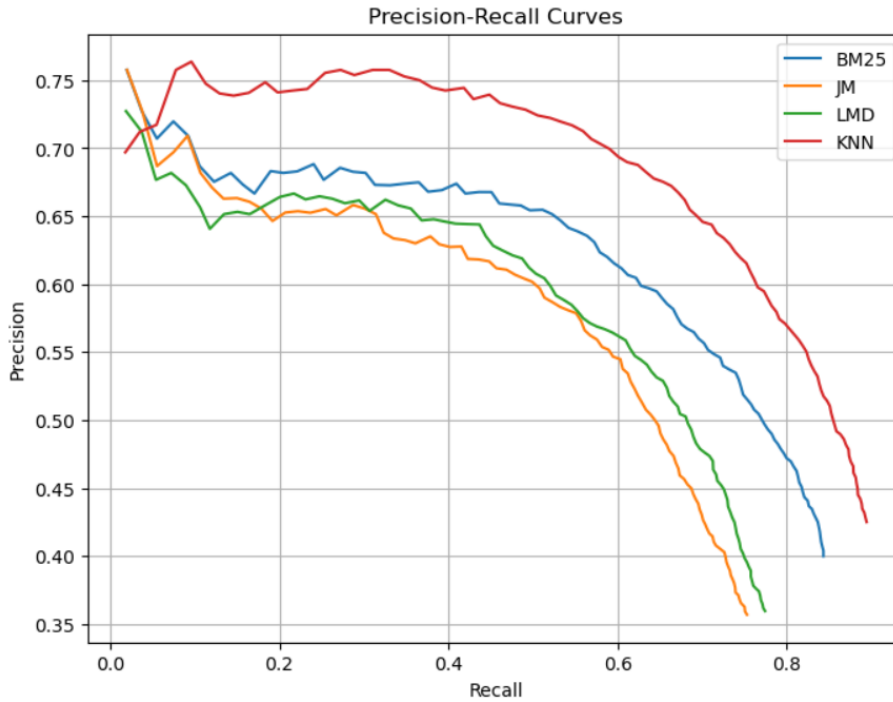


Figure 4: Precision-recall curves for the evaluated retrieval methods.

The precision-recall curves are expected to reflect the same overall tendency observed in the table, with KNN maintaining higher precision across a wider recall range than the lexical baselines.

3.2.4 Query-Level Precision-Recall Analysis

To further analyze system stability, we examine three representative queries: the query with the highest Average Precision (AP), the query with the lowest AP, and a query with median performance.

Figure 5 presents the precision-recall curves for these three queries using the KNN retrieval model.

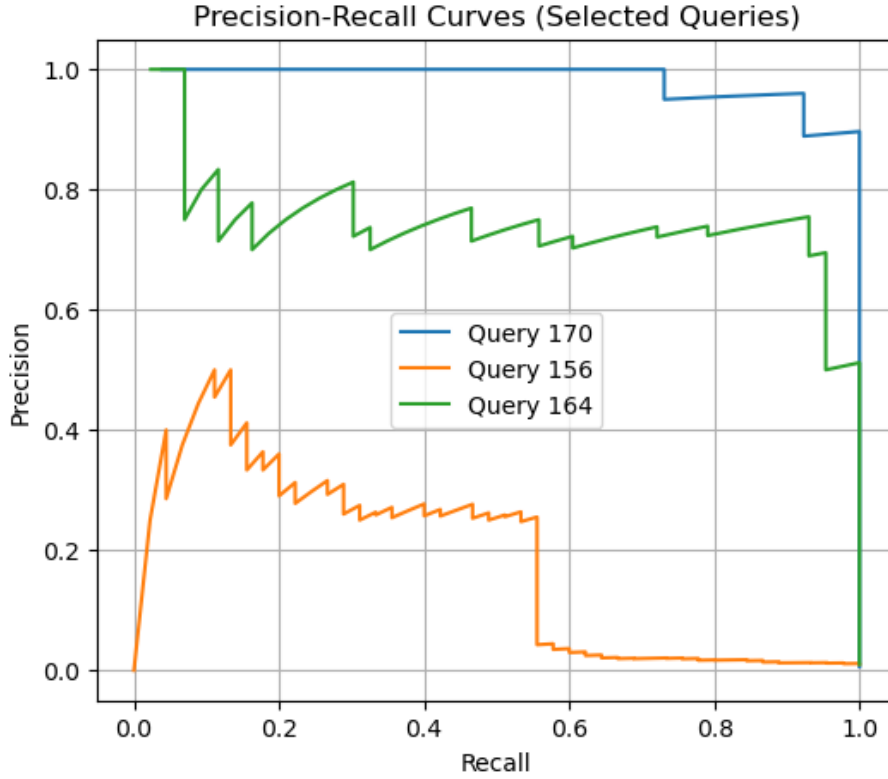


Figure 5: Precision-recall curves for selected queries (best, worst, and median AP).

Topic **170** (best AP) maintains near-perfect precision across most of the recall range, while Topic **156** (worst AP) drops to near-zero precision quickly. Topic **164** shows intermediate behavior with a gradual decline, consistent with vocabulary mismatch being the main failure mode for harder queries.

3.3 Hyperparameter Analysis

The default OpenSearch smoothing parameters ($\lambda = 0.1$, $\mu = 2000$) apply very heavy corpus smoothing. Given our small corpus of 4,194 documents with an average abstract length of 139 words, this is poorly suited — the corpus background is not reliable enough to dominate scoring. We therefore evaluated two alternative configurations on the training split: $\lambda = 0.2/\mu = 150$ and $\lambda = 0.5/\mu = 150$, setting μ near the average document length to balance document and corpus contributions.

Table 2 compares the default and best-tuned configurations for JM and LMD (KNN and BM25 are parameter-free).

Table 2: Default vs. best-tuned smoothing parameters (JM and LMD only).

Method	Config	MAP	NDCG@10	NDCG@100
JM	default ($\lambda = 0.1$)	0.5263	0.6309	0.6976
JM	tuned ($\lambda = 0.5, \mu = 150$)	0.5693	0.6503	0.7404
LMD	default ($\mu = 2000$)	0.5353	0.6144	0.7076
LMD	tuned ($\mu = 150$)	0.5659	0.6315	0.7374

Both methods improve consistently when μ is set near the average document length (139 words), yielding NDCG@10 gains of +3.1% for JM and +2.8% for LMD over the defaults.

3.4 Phase 2 Evaluation

3.4.1 Experimental Setup

The Phase 2 evaluation ran on all 33 test queries using the KNN outputs from Phase 1. Up to 3 reference sentences per retrieved article were selected by MedCPT, yielding 256 judged sentences (5–9 per query). Answers were generated with `google/medgemma-4b-it` and evaluated with GPT-4o through the IAedu API.

3.4.2 Reference Sentence Alignment Results

Table 3 summarizes the LLM-judge labels assigned to the 256 selected reference sentences.

Table 3: LLM-judge reference sentence alignment results over all 33 test queries.

Label	Count	Percentage
Required	60	23.4%
Borderline	42	16.4%
Unnecessary	150	58.6%
Inappropriate	4	1.6%
Total	256	100.0%

The results show that most selected sentences were judged as *Unnecessary*. This does not necessarily mean that the retrieval pipeline failed. The cross-encoder ranks sentences according to query-sentence semantic similarity, while the judge evaluates whether each sentence is necessary for a complete final answer. These are related but not identical objectives. A sentence can be highly related to the topic while still being too general, too methodological, redundant, or not essential for answering the patient’s specific question.

The relatively high number of *Unnecessary* labels therefore indicates that the sentence selection stage has high recall but limited precision. It often retrieves medically related evidence, but also includes background information, study objectives, prevalence statements, or general clinical context that is not needed in the final answer. This is visible, for example, in questions where

the patient asks for a specific practical answer, but the selected references include general disease descriptions or methodological sentences from abstracts.

The *Required* and *Borderline* labels together account for 102 sentences, or 39.8% of all selected evidence. This suggests that the pipeline usually retrieves at least some useful supporting evidence, but the evidence set could be made more focused. A possible improvement would be to reduce the number of selected sentences per article, for example from 3 to 2, or to rank all candidate sentences globally and select only the top reference sentences across all retrieved documents instead of forcing the same number of sentences per article. We attempted to evaluate both alternatives, but the API became unstable during these runs and returned processing errors. The issue also occurred when rerunning the original configuration with the same parameters, suggesting that the failures were related to API availability rather than to the modified selection strategies. For this reason, we report the best complete set of results that we were able to obtain.

Only 4 sentences were labeled *Inappropriate*. These cases generally occurred when the sentence was medically related but potentially misleading or harmful in the context of the patient’s question. For example, evidence about serious complications or severe disease may be inappropriate when the question asks about a minor condition, because it may overstate the risk or fail to answer the intended question.

3.4.3 Answer Entailment Results

Table 4 summarizes the entailment labels assigned to the 33 generated answers.

Table 4: LLM-judge entailment results for generated answers over all 33 test queries.

Entailment Label	Count	Percentage
Supported	11	33.3%
Partially Supported	21	63.6%
Unsupported	1	3.0%
Total	33	100.0%

Most generated answers were judged as *Partially Supported*. This means that the generated answers were generally relevant and grounded in the selected references, but not completely supported by them. Only one answer was judged as *Unsupported*, which suggests that the RAG pipeline was usually able to prevent completely ungrounded generation.

The 11 *Supported* answers show that the system can generate well-grounded responses when the retrieved evidence directly matches the user’s information need. In these cases, the selected references contained concise and answer-relevant sentences, allowing MedGemma to produce an answer that was fully entailed by the evidence.

However, the dominance of *Partially Supported* labels reveals an important limitation. Some answers included details that were present in the retrieved references but not central to the question.

3.4.4 Qualitative Analysis

The Phase 2 results confirm that the overall pipeline is functional: documents retrieved by KNN are converted into sentence-level evidence, the MedCPT cross-encoder ranks the candidate evidence, MedGemma generates citation-bearing answers, and GPT-4o evaluates both evidence alignment and answer entailment.

Nevertheless, the experiment also shows a clear mismatch between cross-encoder relevance and judge necessity. Cross-encoder scores were often extremely high, even for sentences later labeled *Unnecessary*. This suggests that the cross-encoder is effective at identifying topical similarity, but its scores should not be interpreted as calibrated probabilities of answer usefulness. For patient-facing biomedical question answering, a sentence must not only mention related biomedical concepts; it must directly contribute to the answer.

A representative example is the corneal injury query. The selected evidence included the directly useful sentence stating that minor corneal abrasions usually heal within 48 hours, which was judged *Required*. However, other selected sentences described complications, deeper injuries, or general treatment approaches. Although medically related, these sentences were either unnecessary or potentially inappropriate because the patient was asking specifically about the healing time of a minor injury.

Another example is the sleep apnea query. Sentences about complementary and integrative approaches were useful, while sentences about conventional CPAP and oral appliances were judged unnecessary because the patient specifically asked about natural treatments. This shows that the pipeline retrieves semantically related medical evidence, but still needs stronger alignment with the intent expressed in the question and narrative.

3.4.5 Manual Inspection of Judge Labels

We manually inspected two examples from each sentence-level label to check whether the LLM-as-a-judge decisions were reasonable. Overall, the labels were mostly consistent with the prompt definitions.

For *Unnecessary*, the inspected sentences were topically related but did not directly answer the patient question. For example, a CBD sentence described the objective of a study rather than explaining whether CBD affects liver enzymes, and a corneal injury sentence discussed deeper corneal pathology rather than minor injury healing time.

For *Required*, the labels were appropriate because the sentences directly answered the question. The vocal cord paralysis example identified severe airway obstruction requiring emergency tracheostomy, and the spontaneous fracture example explained a relevant cause group for fractures without trauma.

For *Borderline*, the sentences were useful but incomplete. The Afrin/Sudafed example gave context about rhinitis medicamentosa but did not answer the drug-interaction question, while the tennis elbow example stated that the condition is self-limiting but did not provide a treatment recommendation.

For *Inappropriate*, the examples were medically related but potentially misleading or alarming

in the specific patient context. For instance, severe corneal healing complications or facial numbness with sudden hemiparesis may be important warning signs, but they do not directly answer the patient’s original question.

This inspection suggests that the judge labels are mostly coherent. It also confirms that high cross-encoder scores do not always imply answer usefulness, since some *Unnecessary* and *Inappropriate* sentences still received scores close to 1.0.

3.4.6 Limitations and Possible Improvements

The main limitation of the current Phase 2 system is that reference sentence selection is performed independently inside each retrieved article. Since the system selects up to 3 sentences per document, it may keep weak sentences from a lower-quality article while excluding stronger sentences from another article. A better strategy would be to collect all candidate sentences from the top retrieved documents, rank them globally with the cross-encoder, and keep only the top 5 or 6 sentences overall. Like we said before, we attempted to evaluate this alternative, but the API became unstable during the run and returned processing errors.

A second limitation is that the current cross-encoder scores are not calibrated. Many selected sentences receive scores close to 1.0, including sentences later judged unnecessary. Future versions should therefore combine cross-encoder ranking with additional filtering rules, such as removing study-objective sentences, overly generic background sentences, duplicate sentences, and sentences with incomplete biomedical claims.

Overall, the Phase 2 results show that the system is able to generate mostly grounded biomedical answers, with 32 out of 33 answers judged either *Supported* or *Partially Supported*. However, the large proportion of *Unnecessary* reference sentences indicates that the evidence selection stage should be made more selective before the system is used as the basis for the Phase 3 biomedical deep research agent.

4 Conclusion

4.1 Achievements

Phase 1 established the retrieval layer: a PubMed corpus was indexed in OpenSearch and evaluated under BM25, JM, LMD, and KNN. KNN achieved the best results on all test metrics (P@10 0.749, NDCG@10 0.721, MAP 0.672), with BM25 as the strongest lexical baseline. The BERT visualization exercise further illustrated how positional embeddings, contextual layers, and attention heads combine to support cross-encoder relevance scoring.

Phase 2 extended this into a grounded RAG pipeline: MedCPT selected 256 reference sentences, MedGemma generated citation-bearing answers, and GPT-4o judged both evidence alignment and answer entailment. Of 256 sentences, 102 (39.8%) were Required or Borderline; 32 of 33 answers were Supported or Partially Supported, confirming that the pipeline generally keeps generation grounded in the retrieved evidence.

4.2 Limitations

Despite these achievements, the current system still has several limitations. For phase 1 the experiments used a single query formulation based on the concatenation of topic, question, and narrative. Other query formulations, or query expansion strategies, may improve retrieval effectiveness.

In Phase 2, the main limitation was the precision of the reference sentence selection stage. Although the cross-encoder retrieved many topically related sentences, 150 out of 256 selected sentences were labeled *Unnecessary*. This shows that semantic similarity alone is not sufficient for selecting evidence for patient-facing biomedical answers. Some selected sentences were background statements, study objectives, methodological details, duplicated evidence, or medically related statements that did not directly answer the user’s question.

Another limitation is that sentence selection was performed independently within each retrieved article. Since the system selected up to 3 sentences per document, it sometimes retained weak sentences from one article while excluding potentially stronger evidence from another. A more robust approach would be to rank all candidate sentences globally and select only the best evidence across all retrieved documents. We attempted to evaluate this alternative, as well as a configuration with fewer selected sentences per article, but API instability and processing errors prevented us from obtaining complete results for those runs.